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CHAPTER 10

BOILERS, WATER HEATERS AND PRESSURE VESSELS

SECTION MC 1001 GENERAL

1001.1 Scope. This chapter shall establish the minimum safety requirements for and shall govern the installation, alteration and repair of boilers, water heaters and pressure vessels.

Exceptions:

1. Pressure vessels used for unheated water supply.
2. Portable unfired pressure vessels and Interstate Commerce Commission containers.
3. Containers for bulk oxygen and medical gas.
4. Unfired pressure vessels having a volume of 5 cubic feet (0.14 m³) or less operating at pressures not exceeding 250 pounds per square inch (psi) (1723.7 kPa) and located within occupancies of Groups B, F, H, M, R, S and U.
5. Pressure vessels used in refrigeration systems that are regulated by Chapter 11.
6. Pressure tanks used in conjunction with coaxial cables, telephone cables, power cables and other similar humidity control systems.
7. Any boiler or pressure vessel subject to inspection by federal inspectors.

1001.2 Thermal safety (spill) switches. Thermal safety (spill) switches shall be installed in accordance with Section 801.23.3.

1001.3 Clearances. Minimum clearances at boilers, water heaters and pressure vessels shall be maintained in accordance with Chapter 3.

SECTION MC 1002 WATER HEATERS

1002.1 General. Potable water heaters and hot water storage tanks shall be listed and labeled and installed in accordance with the manufacturer's instructions, the *New York City Plumbing Code* and this code. All water heaters shall be capable of being removed without first removing a permanent portion of the building structure. The potable water connections and relief valves for all water heaters shall conform to the requirements of the *New York City Plumbing Code*. Domestic electric water heaters shall comply with UL 174 or UL 1453. Commercial electric water heaters shall comply with UL 1453. Oil-fired water heaters shall comply with UL 732. Approval for oil-fired water heaters 350,000 Btu/h input (103 kW) and above shall be obtained from the New York City Department of Environmental Protection. Thermal solar water heaters shall comply with UL 174 or UL 1453, and Chapter 14 of this code. Heat pump water heaters shall comply with UL 174 or UL 1995.

1002.2 Water heaters utilized for space heating. Water heaters utilized both to supply potable hot water and provide hot water for space-heating applications shall be listed and labeled for such applications by the manufacturer and shall be built in accordance with Section IV of the *ASME Boiler and Pressure Vessel Code* with an "H" code stamp. Water heaters shall be installed in accordance with the manufacturer's instructions, the *ASME Boiler and Pressure Vessel Code* and the *New York City Plumbing Code*. Oil-fired heaters shall comply with UL 732 and shall have an "HW" stamp.

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1002.2.1 Sizing. Water heaters utilized for both potable water heating and space-heating applications shall be sized to prevent the space-heating load from diminishing the required potable water-heating capacity.

1002.2.2 Temperature limitation. Where a combination potable water-heating and space-heating system requires water for space heating, a temperature actuated mixing valve complying with ASSE 1017 shall be provided to temper the water supplied to the potable hot water distribution system in accordance with Section 607 of the *New York City Plumbing Code*.

1002.2.3 Cross-connection. Water heaters utilized for both potable hot water and hot water for space-heating applications shall have separate heating sections and connections for distribution systems and shall not be cross-connected. The potability of the domestic water shall be maintained throughout the system in accordance with the *New York City Plumbing Code*.

1002.3 Supplemental water-heating devices. Potable water-heating devices that utilize refrigerant-to-water heat exchangers shall be approved and installed in accordance with the *New York City Plumbing Code*, the applicable provisions of the *New York City Energy Conservation Code* and the manufacturer's instructions.

SECTION MC 1003 PRESSURE VESSELS

1003.1 General. All pressure vessels shall be constructed and certified in accordance with the *ASME Boiler and Pressure Vessel Code*, and shall bear the label of an approved agency. Pressure vessels shall be installed in accordance with the manufacturer's instructions and nationally recognized standards. Directly fired pressure vessels shall meet the requirements of Section 1004.

1003.2 Piping. All piping materials, fittings, joints, connections and devices associated with systems utilized in conjunction with pressure vessels shall be designed for the specific application and shall comply with the *ASME Boiler and Pressure Vessel Code* and this code.

1003.3 Welding. Welding on pressure vessels shall be performed by welders certified in compliance with nationally recognized standards, including the *National Board Inspection Code* and *ASME Boiler and Pressure Vessel Code*, Section IX. Such welders shall also comply with 12 NYCRR 4-6.2, and 12 NYCRR 14-3.3 through 14-3.18.

SECTION MC 1004 BOILERS

1004.1 Standards. Boilers shall be designed, constructed and certified in accordance with the *ASME Boiler and Pressure Vessel Code*, Section I or IV, and 12 NYCRR Parts 4 and 14. Controls and safety devices for boilers with fuel input ratings of 12,500,000 Btu/h (3663 kW) or less shall meet the requirements of ASME CSD-1. Controls and safety devices for boilers with inputs greater than 12,500,000 Btu/h (3663 kW) shall meet the requirements of NFPA 85. Packaged oil-fired boilers shall be listed and labeled in accordance with UL 726. Packaged electric boilers shall be listed and labeled in accordance with UL 834. Approval for oil-fired boilers 350,000 Btu/h input (102.6 kW) and above shall be obtained from the New York City Department of Environmental Protection.

1004.1.1 Field erected boilers requirements. Field erected boiler installations shall comply with the following:

1. The vessel erector shall hold the appropriate ASME stamp for vessel construction.
2. The licensed installer is responsible for the installation of all controls and burners which shall be designed and constructed in accordance with the requirements of ASME CSD-1 and, as applicable, the *ASME Boiler and Pressure Vessel Code*, 12 NYCRR Parts 4 and 14, and NFPA 85.
3. The New York City Department of Environmental Protection shall approve the installation of the boiler and burner for compatibility, controls and safety devices.
4. Boiler vessels shall comply with one of the following listings:
 - 4.1. UL 726 or UL 795 as applicable; or

4.2. UL 2106.

Exception: Unlisted equipment shall obtain approval from the department.

5. Controls and safety devices shall be tested and inspected in accordance with ASME CSD-1 and, as applicable, the *ASME Boiler and Pressure Vessel Code*, 12 NYCRR Parts 4 and 14, and NFPA 85. Testing and inspection shall be performed by the burner manufacturer's authorized representative in the presence of the licensed installer.
6. Inspection of the installed boiler assembly shall be performed by the department.

1004.2 Installation. In addition to the requirements of this code, the installation of boilers shall conform to the manufacturer's instructions. Operating instructions of a permanent type shall be located in the boiler room and readily accessible. Boilers shall have all controls set, adjusted and tested by the installer. The manufacturer's rating data and the nameplate shall be attached to the boiler.

1004.3 Working clearance. Clearances shall be maintained around boilers, generators, heaters, tanks and related equipment and appliances so as to permit inspection, servicing, repair, replacement and visibility of all gauges. When boilers are installed or replaced, clearance shall be provided to allow access for maintenance and repair per the boiler's listing and manufacturer's instructions. Passageways for inspection around all sides of boilers shall have an unobstructed width of not less than 18 inches (457.2 mm) unless the boiler's listing or department approval or manufacturer's instructions state otherwise.

1004.3.1 Top clearance. Clearances from the tops of boilers to the ceiling or other overhead obstruction shall be in accordance with Table 1004.3.1.

**TABLE 1004.3.1
BOILER TOP CLEARANCES**

BOILER TYPE	MINIMUM CLEARANCES FROM TOP OF BOILER TO CEILING OR OTHER OVERHEAD OBSTRUCTION (feet)
All boilers with manholes on top of the boiler except where a greater clearance is required in this table.	3
All boilers without manholes on top of the boiler except high-pressure steam boilers and where a greater clearance is required in this table.	2
High-pressure steam boilers with steam generating capacity not exceeding 5,000 pounds per hour.	3
High-pressure steam boilers with steam generating capacity exceeding 5,000 pounds per hour.	7
High-pressure steam boilers having heating surface not exceeding 1,000 square feet.	3
High-pressure steam boilers having heating surface in excess of 1,000 square feet.	7
High-pressure steam boilers with input not exceeding 5,000,000 Btu/h.	3

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High-pressure steam boilers with input in excess of 5,000,000 Btu/h.	7
Steam-heating boilers and hot water-heating boilers with input exceeding 5,000,000 Btu/h.	3
Steam-heating boilers exceeding 5,000 pounds of steam per hour.	3
Steam-heating boilers and hot water-heating boilers having heating surface exceeding 1,000 square feet.	3

For SI: 1 foot = 304.8 mm, 1 square foot = 0.0929 m², 1 pound per hour = 0.4536 kg/h, 1 Btu/hr = 0.293 W.

1004.4 Mounting. Equipment and appliances shall be set or mounted on a level base capable of supporting and distributing the weight contained thereon. Boilers, tanks and equipment shall be securely anchored to the structure. Equipment and appliances requiring vibration isolation shall be installed as designed by a registered design professional in accordance with the manufacturer's instructions.

1004.5 Floors. Boilers shall be mounted on floors of noncombustible construction, unless listed for mounting on combustible flooring. Vibration isolation shall be in accordance with Section 313. In buildings with residential occupancies, boilers shall be installed in accordance with the requirements of Section 313.

1004.6 Boiler rooms and enclosures. Boiler rooms and enclosures and access thereto shall comply with Chapters 5 and 10 of the *New York City Building Code* and Chapter 3 of this code. Boiler rooms shall be equipped with a floor drain or other approved means for disposing of liquid waste in accordance with the *New York City Plumbing Code*.

1004.7 Operating adjustments and instructions. Hot water and steam boilers shall have all operating and safety controls set and operationally tested by the installing contractor. A complete control diagram and boiler operating instructions shall be furnished by the installer for each installation. A complete control diagram of a permanent type shall be located in the boiler room and shall be readily accessible.

1004.8 Burner controls. Gas and oil modulating burners shall be provided with burner controls (oil and gas equivalent ratings) in accordance with ASME CSD-1 and NFPA 85.

1004.9 Conversion burners. Conversion burners shall be installed in accordance with ANSI Z21.8.

SECTION MC 1005 BOILER CONNECTIONS

1005.1 Valves. Every boiler or modular boiler shall have a shutoff valve in the supply and return piping. For multiple boiler or multiple modular boiler installations, each boiler or modular boiler shall have individual shutoff valves in the supply and return piping.

Exception: Shutoff valves are not required in a system having a single low-pressure steam boiler of 350,000 Btu/h (103 kW) output or less.

1005.2 Potable water supply. The water supply to all boilers, including but not limited to backflow prevention, shall be connected in accordance with the *New York City Plumbing Code*.

SECTION MC 1006 SAFETY AND PRESSURE RELIEF VALVES AND CONTROLS

1006.1 Safety valves for steam boilers. Steam boilers shall be protected with a safety valve in accordance with the *ASME Boiler and Pressure Vessel Code* or ASME CSD-1.

1006.2 Safety relief valves for hot water boilers. Hot water boilers shall be protected with a safety relief valve in accordance with the *ASME Boiler and Pressure Vessel Code* or ASME CSD-1.

1006.3 Pressure relief for pressure vessels. Pressure vessels shall be protected with a pressure relief valve or pressure-limiting device as required by the manufacturer's instructions for the pressure vessel and in accordance with the *ASME Boiler and Pressure Vessel Code* or ASME CSD-1.

1006.4 Standards of safety and safety relief valves. Safety and safety relief valves shall be listed and labeled, and shall have a minimum rated capacity for the equipment or appliances served. Safety and safety relief valves shall be set at not greater than the nameplate pressure rating of the boiler or pressure vessel and shall be in accordance with the *ASME Boiler and Pressure Vessel Code* or the ASME CSD-1.

1006.5 Installation. Safety or relief valves shall be installed directly into the safety or relief valve opening on the boiler or pressure vessel. Valves shall not be located on either side of a safety or relief valve connection. The relief valve drain shall discharge by gravity to a nonhazardous point of discharge without obstruction.‡

1006.6 Safety and relief valve discharge. Safety and relief valve discharge pipes shall be of rigid pipe that is approved for the temperature of the system. The discharge pipe shall, at a minimum, be the same diameter as the safety or relief valve outlet. Safety and relief valve discharge pipes shall be properly supported to prevent stress on the valve and vessel. Safety and relief valves shall not discharge so as to be a hazard, a potential cause of damage or otherwise a nuisance. Discharge of hazardous materials must be properly contained in a method approved by the department. High-pressure-steam safety valves shall be vented to the outside of the structure. Where a low-pressure safety valve or a relief valve discharges to the drainage system, the installation shall conform to the *New York City Plumbing Code*.

1006.7 Boiler safety devices. Boilers shall be equipped with controls and limit devices as required by the manufacturer's instructions and the conditions of the listing.

1006.8 Electrical requirements. The power supply to the electrical control system shall be from a two-wire branch circuit that has a grounded conductor, or from an isolation transformer with a two-wire secondary. Where an isolation transformer is provided, one conductor of the secondary winding shall be grounded. Control voltage shall not exceed 150 volts nominal, line to line. Control and limit devices shall interrupt the ungrounded side of the circuit. A means of manually disconnecting the control circuit shall be provided and controls shall be arranged so that when deenergized, the burner shall be inoperative. Such disconnecting means shall be capable of being locked in the off position and shall be provided with ready access.

1006.8.1 Remote control (shutdown). A remote control shall be provided to stop the flow of oil and/or gas and combustion air to any burner or fuel-burning internal combustion equipment. Such a control shall be provided along every means of egress from the room in which the burner or equipment is located. Each remote control shall be located outside of, but as close as practicable to the burner or equipment room itself and shall be labeled: "REMOTE CONTROL FOR BURNER."

Exception: Where an outside location is impracticable, the remote control shutdown may be provided immediately inside the room in which the burner or equipment is located. Such location must be accessible at all times.

1006.9 Carbon monoxide detectors. Carbon monoxide detectors shall be provided in all fuel-fired appliance rooms to detect the level of carbon monoxide in the room and signal an alarm. Such detectors shall be listed and installed in accordance with Chapter 9 of the *New York City Building Code*.

SECTION MC 1007 BOILER LOW-WATER CUTOFF

1007.1 General. Steam and hot water boilers shall be protected with dual low-water cutoff control, with each control independently piped to the pressure vessel in accordance with ASME CSD-1. For hydronic boilers, the low-water cut out may be located in the supply piping above the boiler before any intervening valve. A flow-sensing control installed in accordance with ASME CSD-1 shall be considered a low-water cutoff for the purposes of this section.

Exception: Hot water boilers located within a dwelling unit supplying only that unit and having a total heat input of less than 400,000 Btu/h (117.2 kW) may be protected by only one low-water cutoff control.

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1007.2 Operation. Low-water cutoff controls and flow-sensing controls required by Section 1007.1 shall automatically stop the combustion operation of the appliance when the water level drops below the lowest safe water level as established by the manufacturer and in accordance with ASME CSD-1, or when water circulation stops, respectively.

1007.3 Low-water cut out maintenance and testing. Low-water cut outs shall be maintained in accordance with ASME CSD-1 and the manufacturer's specifications. An operator shall test low-water cut outs as follows: (1) high pressure steam boilers every shift, (2) low pressure steam boilers daily and (3) hot water boilers monthly. Slow drain tests shall be conducted for steam boilers every six months. Every year, low-water cut outs and associated piping for steam boilers shall be opened, cleaned and inspected. Records of all testing, cleaning, and inspection required by this section shall be maintained, and made available to the department upon request.

SECTION MC 1008 BOILER BLOWOFF/BLOWDOWN VALVES

1008.1 General. Every boiler shall be equipped with blowoff/blowdown valve(s). The valve(s) shall be installed in the openings provided on the boiler. The minimum quantity and size of the valve(s) and associated piping shall be the quantity and size specified by the boiler manufacturer or the quantity and size of the boiler blowoff/blowdown valve opening. Where the maximum allowable working pressure of the boiler exceeds 15 psig (103.4 kPa), two bottom blowoff valves shall be provided consisting of either two slow-opening valves in series or one quick-opening valve and one slow-opening valve in series, with the quick-opening valve installed closest to the boiler.

1008.2 Discharge. Blowoff/blowdown valves shall discharge to a safe place of disposal. Where discharging to the drainage system, the installation shall conform to the *New York City Plumbing Code*.

1008.3 Maintenance. Blowdown lines shall be inspected annually to verify they are free of deposits and in good working condition.

SECTION MC 1009 HOT WATER BOILER EXPANSION TANK

1009.1 Where required. An expansion tank shall be installed in every hot water system. For multiple boiler installations, not less than one expansion tank is required. Expansion tanks shall be of the closed or open type. Tanks shall be rated for the pressure of the hot water system. Lockable shutoff valves shall be installed at connections to all expansion tanks. Valves shall remain locked in the open position.

1009.2 Closed-type expansion tanks. Closed-type expansion tanks shall be installed in accordance with the manufacturer's instructions. Expansion tanks for systems designed to have an operating pressure in excess of 30 psi (205.8 kPa) shall be constructed and certified in accordance with the *ASME Boiler and Pressure Vessel Code*. The size of the tank shall be based on the capacity of the hot-water-heating system. The minimum size of the tank shall be determined in accordance with the following equation where all necessary information is known:

$$V_t = \frac{(0.00041T - 0.0466)V_s}{\left(\frac{P_a}{P_f}\right) - \left(\frac{P_a}{P_o}\right)} \quad \text{(Equation 10-1)}$$

For SI:

$$V_t = \frac{(0.000738T - 0.03348)V_s}{\left(\frac{P_a}{P_f}\right) - \left(\frac{P_a}{P_o}\right)}$$

where:

V_t = Minimum volume of tanks (gallons) (L).

V_s = Volume of system, not including expansion tanks (gallons) (L).

T = Average operating temperature (°F) (°C).

P_a = Atmospheric pressure (psi) (kPa).

P_f = Fill pressure (psi) (kPa).

P_o = Maximum operating pressure (psi) (kPa).

Where all necessary information is not known, the minimum size of the tank shall be determined from Table 1009.2.

**TABLE 1009.2
CLOSED-TYPE EXPANSION TANK SIZING**

SYSTEM VOLUME IN GALLONS	TANK CAPACITIES IN GALLONS	
	Pressurized Diaphragm Type	Nonpressurized Type
100	9	15
200	17	30
300	25	45
400	33	60
500	42	75
1,000	83	150
2,000	165	300

For SI: 1 gallon = 3.795 L.

1009.3 Open-type expansion tanks. Open-type expansion tanks shall be located not less than 4 feet (1219.2 mm) above the highest heating element. The tank shall be adequately sized for the hot water system. An overflow with a minimum diameter of 1 inch (25.4 mm) shall be installed at the top of the tank. The overflow shall discharge to the drainage system in accordance with the *New York City Plumbing Code*.

SECTION MC 1010 GAUGES

1010.1 Hot water boiler gauges. Every hot water boiler shall have a pressure gauge and a temperature gauge, or a combination pressure and temperature gauge. The gauges shall indicate the temperature and pressure within the normal range of the system's operation.

1010.2 Steam boiler gauges. Every steam boiler shall have a water-gauge glass and a pressure gauge with a siphon. The pressure gauge shall indicate the pressure within the normal range of the system's operation.

1010.2.1 Water-gauge glass. The gauge glass shall be installed in accordance with the *ASME Boiler and Pressure Vessel Code* or ASME CSD-1.

SECTION MC 1011 TESTS

1011.1 Tests. Upon completion of the assembly and installation of boilers and pressure vessels, acceptance tests shall be conducted in accordance with the requirements of the *ASME Boiler and Pressure Vessel Code*. Boilers shall not be placed in operation upon completion of construction until they have been inspected and tested and a certificate of compliance has been issued by the commissioner. All final inspections and tests for boilers shall be witnessed by a qualified boiler inspector in the employ of the department or a duly authorized insurance company as provided in Section 204 of the Labor Law of the State of New York. Equipment having an input of not more than 350,000 Btu/h (103 kW) shall be exempt from this requirement. Where field assembly of pressure vessels or boilers is required, a copy of the completed H-2, P-2 or U-1 Manufacturer's Data Report and the completed Appendix C of ASME CSD-1 required by the *ASME Boiler and Pressure Vessel Code* shall be submitted to the department.

1011.1.1 Fitness for service. When a boiler or pressure vessel has exceeded its useful life or has been subjected to an event that may have impacted the integrity of the pressure vessel, the department may require that a fitness for service study be completed. The study shall be completed in accordance with the *National Board Inspection Code* Part 2, Section 4, and documented in accordance with the National Board 403 form.

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1011.2 Test gauges. An indicating test gauge shall be connected directly to the boiler or pressure vessel where it is visible to the operator throughout the duration of the test. The pressure gauge scale shall be graduated over a range of not less than one and one-half times and not greater than four times the maximum test pressure. Gauges utilized for testing shall be calibrated and certified annually in accordance with ASME B40.100 by the test operator.

†††**1011.3 Periodic boiler inspections.** Periodic boiler inspections shall be performed in accordance with Article 303 of Chapter 3 of Title 28 of the *Administrative Code* and Section 1007.3 of this code. In addition, boiler inspections shall:

1. Be completed in accordance with the *National Board Inspection Code*.
2. Include the review of testing documentation for all controls and safety devices.
3. Verify that the flue connection from the boiler to the chimney is properly sealed and in good working condition.
4. Verify that the combustion air system as originally designed is operational.
5. Verify that the High Pressure Operators' licenses are current and that Low Pressure Operators are qualified per New York State requirements.
6. Include a permanent record of the visit.
7. Be subject to the quality control measures of the department.

1011.4 Pressurized systems containing hazardous materials. Any pressurized system that contains hazardous materials or presents a physical hazard by release shall be pre-approved for installation, subject to testing, and subject to inspection by the department.

1011.5 Nondestructive examination. When required by the department, the nondestructive examination (NDE) requirements, including technique, extent of coverage, procedures, personnel qualification, and acceptance criteria, shall be performed in accordance with the applicable provisions of the edition of the code of construction for the pressure-vessel in effect at the time of installation. Weld repairs and alterations shall be subject to the same nondestructive examination requirements. Where this is not possible or practicable, alternative NDE methods acceptable to the inspector and the department may be used.

SECTION MC 1012 MAXIMUM TEMPERATURE

1012.1 Maximum temperature. Maximum indoor temperature in spaces surrounding boilers, water heaters, and pressure vessels shall not exceed the operational temperature of the installed equipment or 104°F (40°C).